BrandGuard AI – Intelligent Brand Governance Platform

Inkloom AI Hackathon · Round 2 (The Build) · October 2026

Project: CloudNova BrandGuard AI	**Stack:** React 18 · Flask 3 · Gemini 2.5 Flash
GitHub: https://github.com/bunnysai872-crypto/BRANDGUARD-AI	**Live Demo:** https://brandguard-ai-tau.vercel.app/login
Backend API: https://brandguard-ai-fdfl.onrender.com	**Health Check:** https://brandguard-ai-fdfl.onrender.com/api/health
Tests: 39 / 39 Passed · pytest 9.1	**Build:** Vite 5 · 0 errors · 0 warnings

1. Problem Statement

Enterprise marketing teams and creative agencies face acute brand safety risks when generating high-velocity promotional content. The proliferation of generative AI tools has accelerated output tenfold but dramatically elevated the risk of unsubstantiated claims (*'100% downtime-proof'*, *'zero cost forever'*), off-brand tone deviations, and hallucinated organizational facts. Manual review cycles in spreadsheets lack audit history, versioning snapshots, and objective rule enforcement, exposing organizations to FTC, ASA, and SEC advertising violations.

2. Proposed Solution

BrandGuard AI is a multi-tenant brand governance platform delivering an end-to-end human-in-the-loop compliance workflow:

- **Centralized Brand Rulebooks:** Full governance across 10 categories — Tone, Terminology, Messaging, Approved Facts, Restricted Claims, Prohibited Claims, Mandatory Disclaimer, Visual Identity, Accessibility, Other.
- **Evidence-Grounded AI Auditing:** Google Gemini 2.5 Flash evaluates copy strictly against active guidelines, returning structured JSON findings with severity (Critical/High/Medium/Low), guideline citations (DB-validated to prevent hallucination), suggested rewrite, and confidence level.
- **Immutable Revision Snapshots:** Every content edit creates an unalterable ContentRevision record. Stale reports are automatically flagged (is_outdated=True), blocking approval.
- **Manager Sign-Off & Immutable Audit Trail:** Content is locked during review. Managers approve or request changes with mandatory comments; all decisions are permanently logged in ReviewDecision with reviewer identity and timestamp.
- **In-App Publishing:** Brand Managers publish Approved content; published_at and published_by_id are persisted. Published submissions are permanently locked.

3. Completed & Verified Features

Phase	Feature Area	Key Capabilities
1	**Auth & RBAC**	Flask-JWT-Extended · PBKDF2 hashing · admin/creator roles · ProtectedRoute SPA guards · idempotent demo-login endpoint
2	**Brand Governance**	Brand profiles · 10-category guideline taxonomy · severity levels · team membership · cross-brand data isolation · file text extraction (pypdf)
3	**Gemini AI Engine**	google-genai SDK · gemini-2.5-flash · JSON output mode · guideline ID validation · stale report detection · malformed response handling

4	**Manager Workflow**	Approval queue · high-severity acknowledgement modal · Approved/Changes Requested decisions · ReviewDecision audit timeline · status locking
5	**Publishing & Analytics**	POST /publish endpoint · Published archive UI · governance analytics (severity dist., category dist., common issues, timeline trends) · Vercel proxy fix

4. Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite 5	SPA, HMR, role-based ProtectedRoute, Tailwind CSS
Backend	Python 3.12 + Flask 3.0	Application Factory, modular blueprints, RESTful API
Database	SQLAlchemy 3.1 / PostgreSQL	10-table relational schema; SQLite for local dev
Auth	Flask-JWT-Extended 4.6	Stateless JWT, HMAC-SHA256, 7-day TTL, role claims
AI Engine	google-genai / Gemini 2.5 Flash	JSON output mode, temp=0.1, structured findings
WSGI	Gunicorn 21.2	Multi-worker production server (Render)
Hosting	Vercel + Render	Vercel (SPA + API proxy) · Render (Flask + PostgreSQL)
Testing	pytest 9.1	39 integration tests, in-memory SQLite fixtures

5. Automated Test Coverage — 39 / 39 Passed

All tests run with *pytest 9.1* against an in-memory SQLite database (no live API calls for AI tests — Gemini is mocked). Zero failures on 2026-10-24.

Test File	Tests	Scope	Result
test_auth.py	11	Registration · login · JWT · RBAC · demo-login idempotency	✓ PASS
test_hackathon_upgrades.py	9	Guideline fields · cross-brand isolation · PDF extraction · analytics · publish security	✓ PASS
test_phase2.py	3	Brand management · guideline CRUD · revision history	✓ PASS
test_phase3.py	5	Gemini API key missing · mocked analysis · ID sanitization · stale detection · malformed JSON	✓ PASS
test_phase4.py	2	Approval validation · manager review lifecycle	✓ PASS
test_production_workflows.py	3	Health check safety · cross-user isolation · full E2E lifecycle	✓ PASS
test_publishing.py	6	Manager-only publish · creator blocked · non-Approved blocked · cross-brand isolation · edit lock	✓ PASS

6. Progress from Round 1 to Round 2

• **Complete Database Schema:** 10-table relational SQLAlchemy schema with cascading deletes, indexed FK, and non-destructive auto-migration.

• **Gemini AI Integration:** google-genai SDK, gemini-2.5-flash, JSON output mode (temp=0.1), structured findings with guideline ID validation.

• **Immutable Revision History:** Every content edit snapshots an unalterable ContentRevision; stale-report detection blocks outdated approvals.

• **Publishing Workflow:** POST /api/submissions/:id/publish — Brand Manager only, Approved status only; permanent edit lock post-publish.

• **Governance Analytics:** Real-time DB-computed dashboard (status counts, severity dist., common issues, timeline trends).

• **39 Automated Tests:** Full pytest coverage across auth, AI compliance, manager workflow, publishing, and cross-user isolation.

• **Live Deployment:** Vercel frontend + Render backend with PostgreSQL; 405 POST bug fixed via API proxy rewrite in vercel.json.

7. Planned Round 3 Improvements

• **Visual Brand Compliance Engine:** Multimodal Gemini analysis of image creatives (PNG/JPG banners, PDF collateral) — logo variant detection, color palette validation, disclaimer legibility. Findings stored separately from text findings.

• **Multi-Format & Accessibility Review:** HTML email, social post, and document format analysis; reading level estimation; alt-text presence checks; platform character-limit validation.

• **Brand Governance Intelligence:** Per-creator compliance scoring, recurring violation detection, guideline effectiveness signals — all computed from existing DB records, no new schema required.

8. Known Limitations

• **In-App Publishing Only:** 'Publish' records status in the database; no external CMS, social media, or ad network integration.

• **Text Content Only:** AI compliance analysis is text-only; images, video, and audio are not analyzed.

• **Scanned PDF / OCR:** pypdf extracts embedded text only; scanned image-based PDFs return empty or incomplete extraction.

• **Demo Login Security:** /api/auth/demo-login auto-creates accounts for evaluator convenience; not production-grade.

• **Render Cold Start:** Render free tier may take 30–60 seconds to respond after inactivity.

9. Judge / Evaluator Quickstart

Live Demo: Open https://brandguard-ai-tau.vercel.app/login
Click **'Content Creator'** or **'Brand Manager'** in the One-Click Login panel (auto-creates demo accounts, no manual seeding required).

Demo Credentials (auto-created on first use):
• Brand Manager: *manager@brandguard.ai* / *Password123!*
• Content Creator: *creator@brandguard.ai* / *Password123!*

Or seed realistic data (run *python seed_demo.py* in BACKEND):
• judge.manager@brandguard.ai / DemoPassword123!
• judge.creator@brandguard.ai / DemoPassword123!

10. Third-Party & AI Assistance Disclosure

AI SDK: Official google-genai Python SDK (Apache 2.0). No API keys are committed to the repository.
Core Libraries: Flask, Flask-SQLAlchemy, Flask-JWT-Extended, Flask-CORS, Werkzeug, python-dotenv, psycopg2-binary, psycopg[binary], pypdf, Gunicorn, React 18, Vite 5, Tailwind CSS, Lucide React, Axios, ReportLab.
AI Coding Assistance: Google Antigravity IDE (Gemini 2.5) was used as an agentic pair-programmer for architecture, code generation, test writing, and documentation. All code validated against automated pytest suites and production build tools.